Cheela Anvesh

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github.com/AnveshCheela | Portfolio

Education

Degree/Certificate	Institute	CGPA/Percentage	Year
B.Tech., CSE	Indian Institute of Information Technology, Sricity	7.3 (upto 5th Sem)	2027
Senior Secondary	Turito College	975/1000	2023
Secondary	Vagdevi Vidyalayam High School	10 CGPA	2021

Projects

GymRats – Comprehensive Fitness Management Platform *GitHub Link*

Tools: MongoDB, Express.js, React.js, Node.js, Redis, Docker, GitHub Actions

- Architected a full-stack MERN platform with role-specific dashboards for Members, Trainers, and Administrators across 30+ React pages and 80+ RESTful API endpoints, consolidating fragmented gym operations into a single data-driven system.
- Engineered JWT (24h expiry) and Google OAuth 2.0 authentication with role-based access control middleware, managing tiered memberships (Basic/Gold/Platinum) with automated renewal, expiry tracking, and payment recording across 16 MongoDB collections.
- Delivered end-to-end fitness features – personalized workout tracking, nutrition macro logging, trainer assignment, and appointment scheduling – structured using a 5-bounded-context Domain-Driven Design with 11 aggregates, 16 entities, and 23 value objects.
- Boosted backend performance using Redis TTL caching (300–3600s) on high-frequency admin endpoints, MongoDB full-text indexes, and float32 memory optimization; deployed via Docker on Render with Vercel CDN for frontend delivery.
- Automated quality assurance with a CI/CD pipeline via GitHub Actions that runs 88 Jest test cases, ESLint validation, and Vite production builds on every push, achieving zero broken deployments to production.

OpsGuardian – Automated AI-Powered Incident Management Platform *GitHub Link*

Tools: Next.js, React, Tailwind CSS, Node.js, Express, TypeScript, PostgreSQL, Prisma ORM, Redis, BullMQ, WebSockets (Socket.io), AWS S3, Google Gemini AI, Resend API

- Architected a multi-tenant, event-driven incident management SaaS platform that automatically ingests, deduplicates, and escalates system crashes in real-time.
- Engineered an asynchronous processing pipeline using Node.js, Redis, and BullMQ to decouple heavy AI and cloud tasks from the main thread, resulting in zero-blocking HTTP webhook ingestion.
- Implemented a custom Idempotency Middleware using SHA-256 hashing and Redis TTL to instantly drop duplicate crash payloads, preventing alert fatigue for on-call engineers.
- Integrated the Google Gemini AI API and AWS S3 into background workers to perform automated Root Cause Analysis (RCA) on raw stack traces, drastically reducing mean-time-to-resolution (MTTR).
- Developed an automated Escalation Engine using the Resend HTTP API that schedules delayed background jobs and recursively traverses company hierarchies to alert the next available engineer if SLAs are missed.
- Built a real-time, responsive frontend dashboard using Next.js 15, React, and Tailwind CSS, leveraging Socket.io WebSockets to instantly broadcast state changes (like incident creation and acknowledgment) to specific team rooms.

VerifyFlow – Digital KYC Verification Platform *GitHub Link*

Tools: React.js, Node.js, Express.js, PostgreSQL, Redis, BullMQ, JWT, Tesseract.js (OCR), Docker, Resend API

- Architected a full-stack KYC verification platform automating identity document review for digital-banking-style onboarding, decoupling OCR and fraud-checking from the request thread via a Redis-backed BullMQ job queue.
- Engineered an OCR extraction pipeline using Tesseract.js to parse uploaded PAN/Aadhaar documents and extract structured fields (Name, DOB, ID Number), with validation and retry handling for low-quality scans.
- Implemented a rule-based Fraud Detection Engine flagging submissions on name mismatches, duplicate document hashes, and invalid ID formats to reduce manual reviewer load on clean cases.
- Built JWT-based authentication with role-based access control (User/Admin) and audit-logging middleware recording every state-changing action across the verification lifecycle for compliance traceability.
- Developed an asynchronous notification service via the Resend API, triggering BullMQ delayed jobs to email users on verification status changes.
- Designed a React-based Admin Review Dashboard for real-time queue management, with Redis caching on high-frequency status/queue endpoints to minimize repeat-read latency – fully containerized and deployed via Docker Compose.

Skills

- **Programming Languages:** C++, Java, Python, JavaScript, TypeScript, SQL
- **Data Structures & Algorithms:** Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Graphs, Hashing, Recursion, Dynamic Programming
- **Object-Oriented Programming:** Encapsulation, Inheritance, Polymorphism, Abstraction, SOLID Principles
- **Backend & Architecture:** Node.js, Express.js, REST APIs, WebSockets (Socket.io), Microservices, Event-Driven Architecture, Message Queues (BullMQ), Authentication (JWT), Role-Based Access Control
- **Databases & ORMs:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis (Caching & Queues), Prisma ORM (Indexing, Schema Design)
- **Web Technologies:** React.js, Next.js, Tailwind CSS, HTML, CSS
- **Cloud, DevOps & AI:** AWS (Lambda, S3, SQS, API Gateway, CloudWatch), Docker, Git, GitHub Actions, Vercel, Railway, Generative AI Integration (Google Gemini)

Certifications

- AWS Cloud Practitioner Essentials

Achievements

- Solved 150+ Data Structures and Algorithms problems on LeetCode.
- Built and deployed multiple full-stack and cloud-native applications using MERN and AWS technologies.